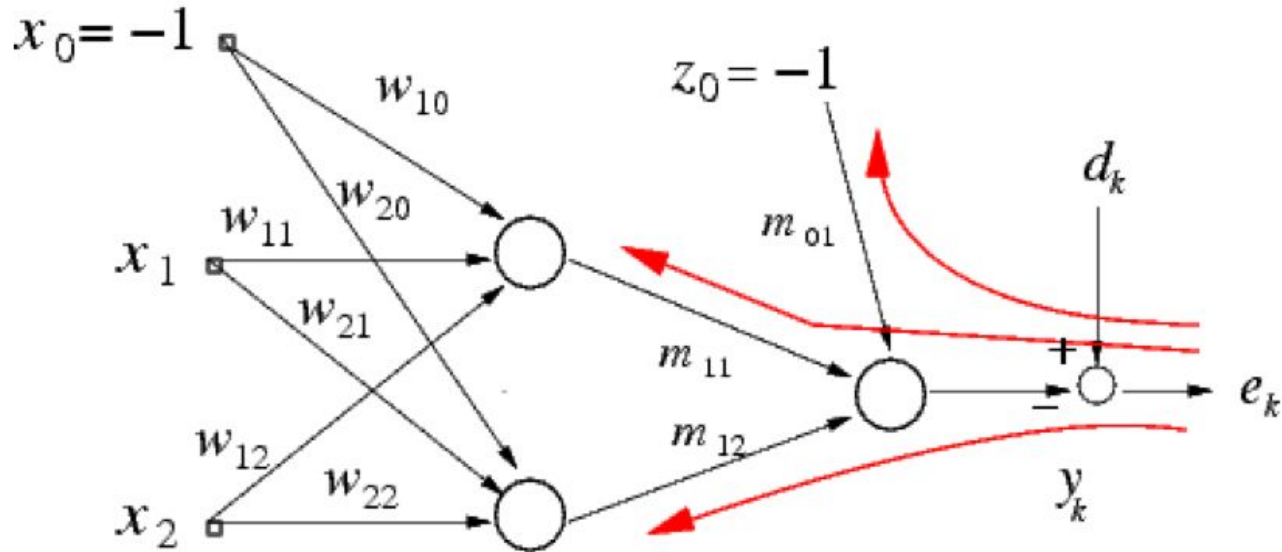




Aprendizagem e camadas ocultas

Prof. Elder Santos

Backpropagation MLP (Multi-Layered Perceptron)



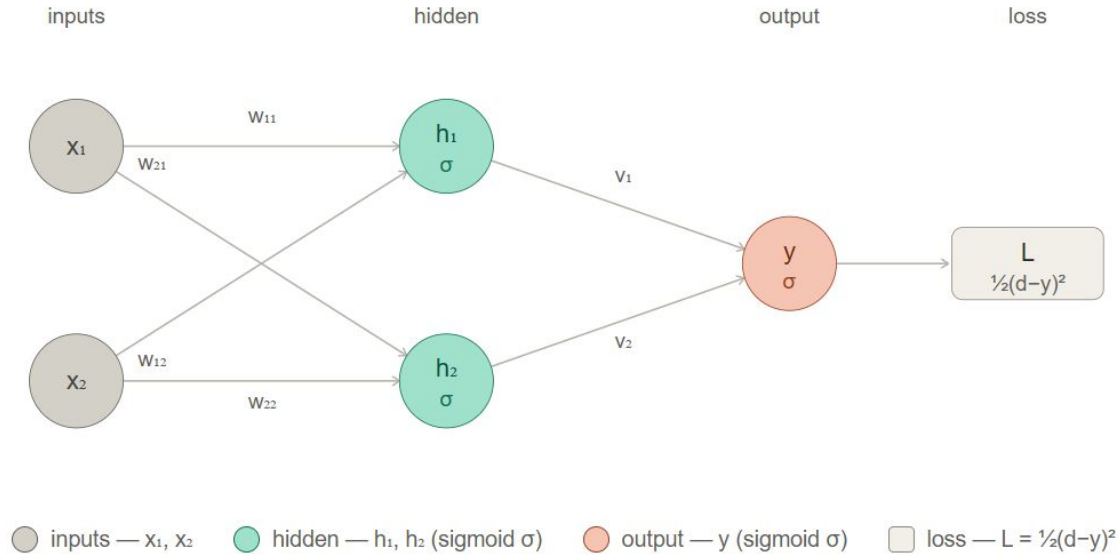


Superfície Erro X Pesos



Gradiente

Função de perda / erro

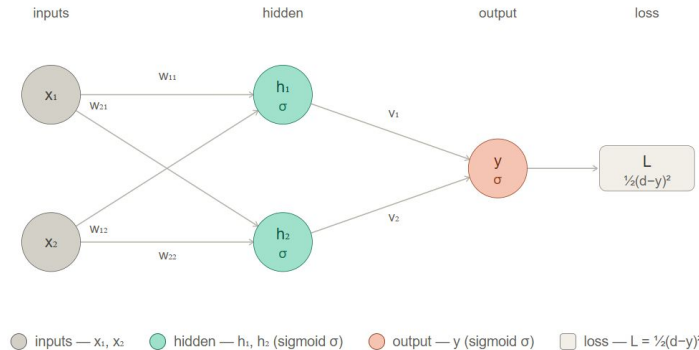


$$L(w_{11}, w_{12}, w_{21}, w_{22}, v_1, v_2) = \frac{1}{2} \cdot (d - \sigma(v_1 \cdot \sigma(w_{11}x_1 + w_{12}x_2) + v_2 \cdot \sigma(w_{21}x_1 + w_{22}x_2)))^2$$

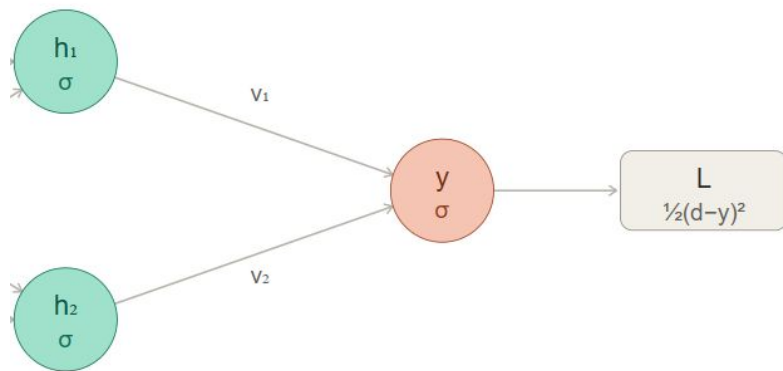
Gradientes

$$L(w_{11}, w_{12}, w_{21}, w_{22}, v_1, v_2) = \frac{1}{2} \cdot (d - \sigma(v_1 \cdot \sigma(w_{11} x_1 + w_{12} x_2) + v_2 \cdot \sigma(w_{21} x_1 + w_{22} x_2)))^2$$

$$\nabla L = (\partial L / \partial v_1, \partial L / \partial v_2, \partial L / \partial w_{11}, \partial L / \partial w_{12}, \partial L / \partial w_{21}, \partial L / \partial w_{22})$$



O primeiro (último) gradiente



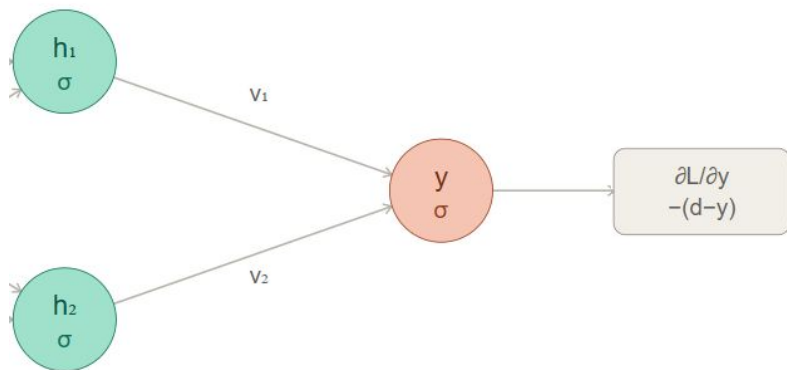
$$L = \frac{1}{2} \cdot (d - y)^2 \quad (\text{perda/erro})$$

$$y = \sigma(z_{\text{out}}) \quad (\text{função de ativação})$$

$$z_{\text{out}} = v_1 \cdot h_1 + v_2 \cdot h_2 \quad (\text{ativação de } y)$$

$$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial z_{\text{out}}} \cdot \frac{\partial z_{\text{out}}}{\partial v_1}$$

O primeiro (último) gradiente - $\partial L / \partial y$



$$\partial L / \partial v_1 = \partial L / \partial y \cdot \partial y / \partial z_{\text{out}} \cdot \partial z_{\text{out}} / \partial v_1$$

$$L = \frac{1}{2} \cdot (d - y)^2$$

$$L = \frac{1}{2} \cdot u^2$$

$$dL/du = \frac{1}{2} \cdot 2 \cdot u^1 = u = d - y$$

$$\partial(d - y) / \partial y = \partial d / \partial y - \partial y / \partial y$$

$$\partial d / \partial y = 0 \quad \partial y / \partial y = 1 \quad \partial(d - y) / \partial y = 0 - 1 = -1$$

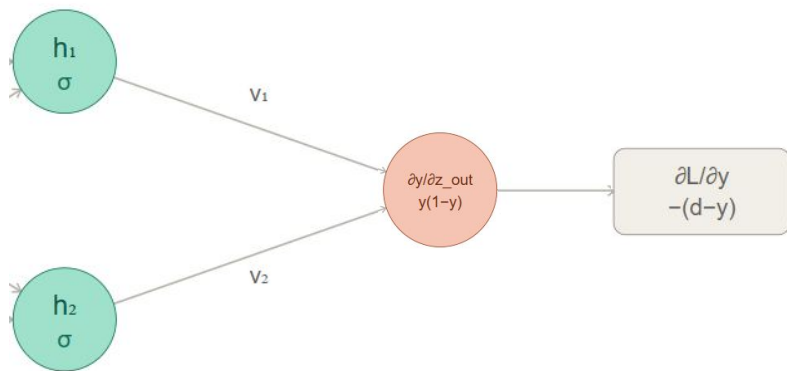
$$\partial L / \partial y = \frac{1}{2} \cdot 2 \cdot (d - y) \cdot \partial(d - y) / \partial y$$

$$= (d - y) \cdot (-1)$$

$$= -(d - y)$$

$$= -e$$

O primeiro (último) gradiente - $\partial y / \partial z_{\text{out}}$



$$\partial L / \partial v_1 = \partial L / \partial y \cdot \partial y / \partial z_{\text{out}} \cdot \partial z_{\text{out}} / \partial v_1$$

$$y = \sigma(z_{\text{out}}) = 1 / (1 + e^{-(z_{\text{out}})})$$

$$\begin{aligned} \partial y / \partial z_{\text{out}} &= \sigma(z_{\text{out}}) \cdot (1 - \sigma(z_{\text{out}})) \\ &= y \cdot (1 - y) \end{aligned}$$

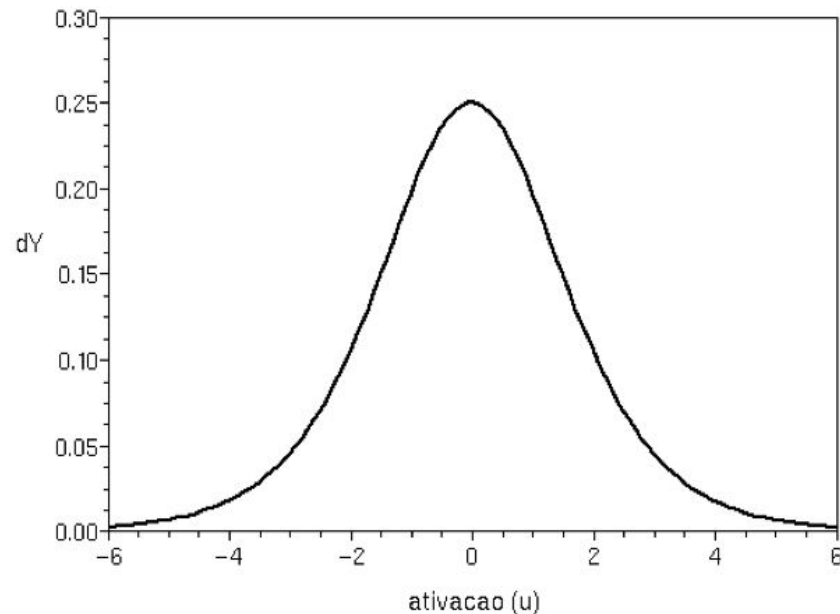
a própria derivada da sigmoide

Função de Ativação MLP

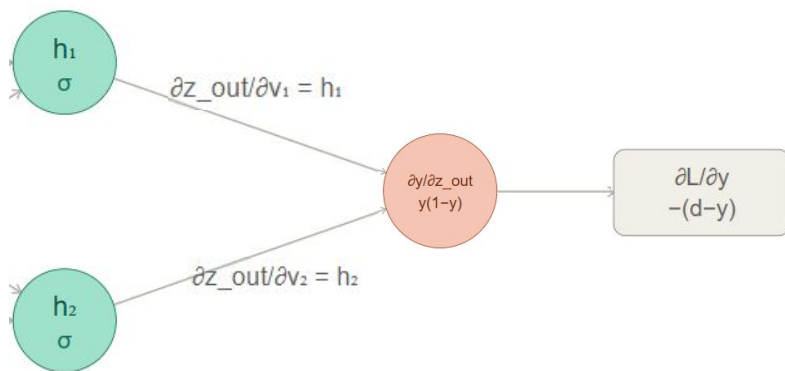
derivada da
sigmoide logística

$$y'_i(t) = \frac{d y_i(t)}{d u_i(t)}$$

$$y'_i(t) = y_i(t)[1 - y_i(t)]$$



O primeiro (último) gradiente - $\partial z_{\text{out}}/\partial v_1$



$$\partial L/\partial v_1 = \partial L/\partial y \cdot \partial y/\partial z_{\text{out}} \cdot \partial z_{\text{out}}/\partial v_1$$

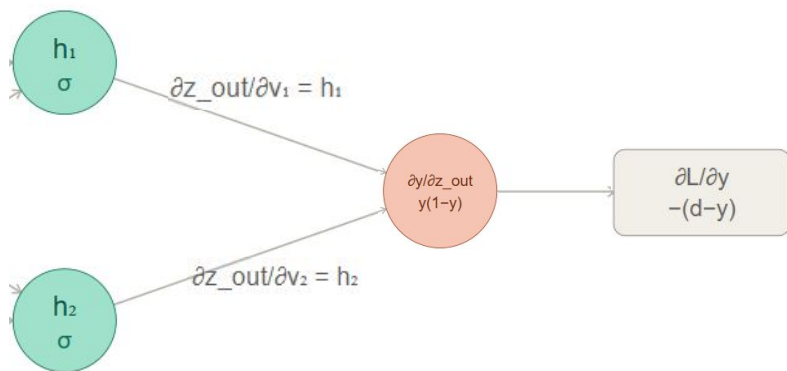
$$z_{\text{out}} = v_1 \cdot h_1 + v_2 \cdot h_2$$

$$\partial z_{\text{out}}/\partial v_1 = \partial(v_1 \cdot h_1)/\partial v_1 + \partial(v_2 \cdot h_2)/\partial v_1$$

$$= h_1 + 0$$

$$= h_1$$

O primeiro (último) gradiente - $\partial z_{\text{out}}/\partial v_1$



$$\partial L/\partial v_1 = \partial L/\partial y \cdot \partial y/\partial z_{\text{out}} \cdot \partial z_{\text{out}}/\partial v_1$$

$$= (-e) \cdot y(1-y) \cdot h_1$$

$$\partial L/\partial v_1 = (-0.949) \cdot (0.051 \times 0.949) \cdot 0.731$$

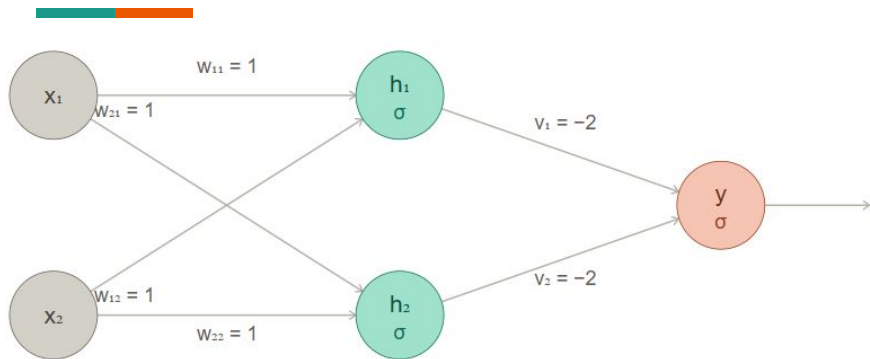
$$= (-0.949) \cdot 0.048 \cdot 0.731$$

$$= -0.034$$

$$\delta_{\text{out}} = -e \cdot y(1-y)$$

$$\partial L/\partial v_2 = \delta_{\text{out}} \cdot h_2$$

Gradiente descendente (em v_1 e v_2)



$$\begin{aligned}\partial L / \partial v_1 &= (-0.949) \cdot (0.051 \times 0.949) \cdot 0.731 \\ &= (-0.949) \cdot 0.048 \cdot 0.731 \\ &= -0.034\end{aligned}$$

$$\begin{aligned}v_1 &\leftarrow v_1 - \eta \cdot \partial L / \partial v_1 \\ &= -2 - 0.5 \cdot (-0.034) \\ &= -2 + 0.017 \\ &= -1.983\end{aligned}$$

*** O gradiente descendente é aplicado apenas no final!**

Gradiente W

$L \rightarrow y \rightarrow z_{\text{out}} \rightarrow h_1 \rightarrow z_1 \rightarrow w_{11}$

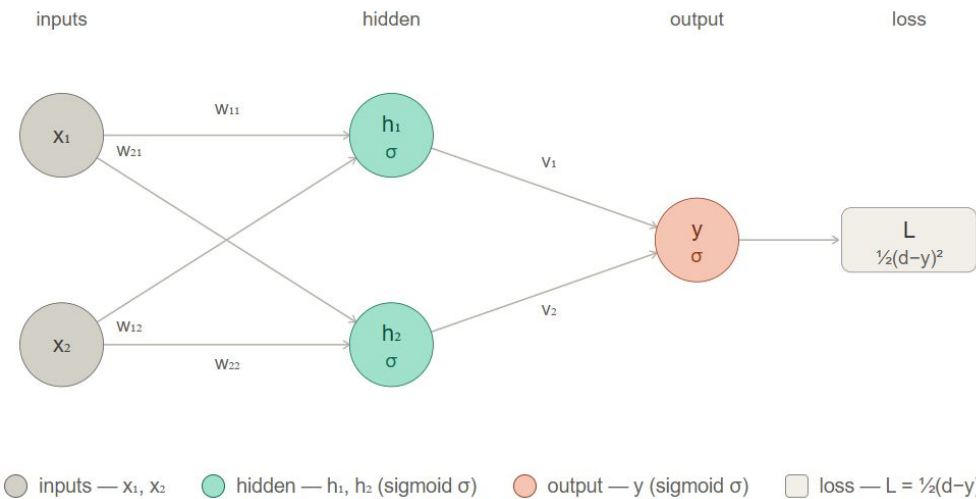
$$\partial L / \partial w_{11} = \partial L / \partial y \cdot \partial y / \partial z_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$\partial L / \partial w_{11} = \delta_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$\delta_{\text{out}} = (-e) \cdot y(1-y)$$

$$= (-0.949) \cdot 0.048$$

$$= -0.046$$



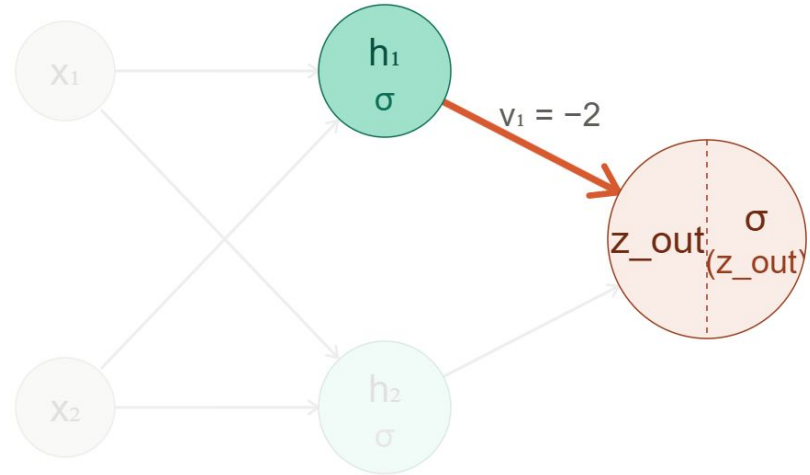
Gradiente W - $\partial z_{\text{out}} / \partial h_1$

$$\partial L / \partial w_{11} = \delta_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$z_{\text{out}} = v_1 \cdot h_1 + v_2 \cdot h_2$$

$$\partial z_{\text{out}} / \partial h_1 = v_1$$

$$v_1 = -2$$





Gradiente W - $\partial h_1 / \partial z_1$

$$\partial L / \partial w_{11} = \delta_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$h_1 = \sigma(z_1)$$

$$\partial h_1 / \partial z_1 = h_1(1-h_1)$$

$$= 0.731 \times 0.269$$

$$= 0.197$$



Gradiente W - $\partial z_1 / \partial w_{11}$

$$\partial L / \partial w_{11} = \delta_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$z_1 = w_{11}x_1 + w_{12}x_2$$

$$\partial z_1 / \partial w_{11} = x_1 = 1$$



Gradiente W_{11}

$$\partial L / \partial w_{11} = \delta_{\text{out}} \cdot \partial z_{\text{out}} / \partial h_1 \cdot \partial h_1 / \partial z_1 \cdot \partial z_1 / \partial w_{11}$$

$$= \delta_{\text{out}} \cdot v_1 \cdot h_1(1-h_1) \cdot x_1$$

$$= (-0.046) \cdot (-2) \cdot (0.197) \cdot (1)$$

$$= +0.018$$



Gradiente descendente w_{11}

$$w_{11} \leftarrow w_{11} - \eta \cdot \partial L / \partial w_{11}$$

$$= 1 - 0.5 \cdot (0.018)$$

$$= 0.991$$

*** O gradiente descendente é aplicado apenas no final!**

Backpropagation - Forward pass

$$z_1 = w_{11} \cdot x_1 + w_{12} \cdot x_2$$

$$h_1 = \sigma(z_1)$$

$$z_2 = w_{21} \cdot x_1 + w_{22} \cdot x_2$$

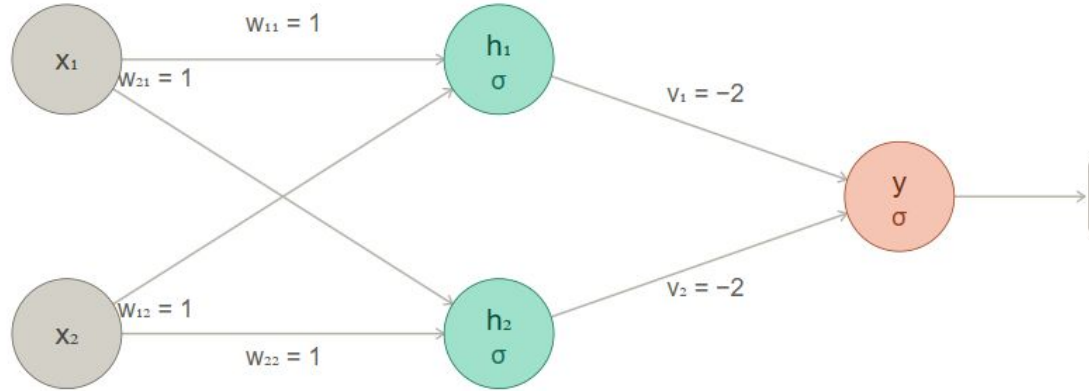
$$h_2 = \sigma(z_2)$$

$$z_{\text{out}} = v_1 \cdot h_1 + v_2 \cdot h_2$$

$$y = \sigma(z_{\text{out}})$$

$$e = d - y$$

$$L = \frac{1}{2} \cdot e^2$$



Backpropagation - Backward pass



$\delta_{\text{out}} = -e \cdot y(1-y) \leftarrow$ delta da saída, reutilizado abaixo

[gradientes camada de saída]

$$\partial L / \partial v_1 = \delta_{\text{out}} \cdot h_1$$

$$\partial L / \partial v_2 = \delta_{\text{out}} \cdot h_2$$

[deltas da camada oculta]

$$\delta_{h_1} = \delta_{\text{out}} \cdot v_1 \cdot h_1(1-h_1) \leftarrow \text{considera } v_1 \text{ antigo}$$

$$\delta_{h_2} = \delta_{\text{out}} \cdot v_2 \cdot h_2(1-h_2) \leftarrow \text{considera } v_2 \text{ antigo}$$

[gradientes da camada oculta]

$$\partial L / \partial w_{11} = \delta_{h_1} \cdot x_1$$

$$\partial L / \partial w_{12} = \delta_{h_1} \cdot x_2$$

$$\partial L / \partial w_{21} = \delta_{h_2} \cdot x_1$$

$$\partial L / \partial w_{22} = \delta_{h_2} \cdot x_2$$

Backpropagation - Atualização dos pesos

Gradiente descendente

$$v_1 \leftarrow v_1 - \eta \cdot \partial L / \partial v_1$$

$$v_2 \leftarrow v_2 - \eta \cdot \partial L / \partial v_2$$

$$w_{11} \leftarrow w_{11} - \eta \cdot \partial L / \partial w_{11}$$

$$w_{12} \leftarrow w_{12} - \eta \cdot \partial L / \partial w_{12}$$

$$w_{21} \leftarrow w_{21} - \eta \cdot \partial L / \partial w_{21}$$

$$w_{22} \leftarrow w_{22} - \eta \cdot \partial L / \partial w_{22}$$